

Heaven's Light is Our Guide



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Chapter 2 ~ Mathematical Tools in Digital Image Processing

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Array versus Matrix Operation

- ❖ Images can be viewed as array of pixels.
- ❖ An array operation involves one or more images is carried out on a pixel by pixel basis.
- ❖ Images can also be viewed equivalently as matrices.
- ❖ There are many situations in which operations between images are carried out using matrix theory.

Array versus Matrix Operation

- For example, consider the following 2x2 images:

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

The *array product* of these two images is

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{12}b_{12} \\ a_{21}b_{21} & a_{22}b_{22} \end{bmatrix}$$

On the other hand, the *matrix product* is given by

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

Array versus Matrix Operation

❖ We assume **array operations** throughout the course:

- For example, consider the following 2x2 images: $I = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

For example,

❖ When we refer to raising an image to a power, we mean that each individual pixel is raised to that power;

- Power of an image: $I^2 = \begin{bmatrix} a_{11}^2 & a_{12}^2 \\ a_{21}^2 & a_{22}^2 \end{bmatrix}$

❖ When we refer to dividing an image by another, we mean that the division is between corresponding pixel pairs, and so on.

- Divide an image by another: $I/I = \begin{bmatrix} a_{11}/a_{11} & a_{12}/a_{12} \\ a_{21}/a_{21} & a_{22}/a_{22} \end{bmatrix}$

Linear versus Nonlinear Operation

✓ Consider a general operator, H , that produces an output image, $g(x,y)$, for a given input image, $f(x,y)$:

$$H[f(x, y)] = g(x, y)$$

✓ H is said to be a linear operator if,

$$\begin{aligned} H[a_i f_i(x, y) + a_j f_j(x, y)] &= a_i H[f_i(x, y)] + a_j H[f_j(x, y)] \\ &= a_i g_i(x, y) + a_j g_j(x, y) \end{aligned}$$

Where, a_i and a_j are arbitrary constant and $f_i(x, y)$ and $f_j(x, y)$ are two images of same sizes.

Linear versus Nonlinear Operation

Let's consider the general operator 'H' as a summation operator 'Σ'

$$\begin{aligned}\Sigma[a_i f_i(x, y) + a_j f_j(x, y)] &= \Sigma a_i f_i(x, y) + \Sigma a_j f_j(x, y) \\ &= a_i \Sigma f_i(x, y) + a_j \Sigma f_j(x, y) \\ &= a_i g_i(x, y) + a_j g_j(x, y)\end{aligned}$$

Linear versus Nonlinear Operation

Given that, $f_1 = \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix}$ and $f_2 = \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix}$ where, $a_1 = 1, a_2 = -1$

$$\text{Now, } \Sigma[a_i f_i(x, y) + a_j f_j(x, y)] = \Sigma \left[1 * \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} + (-1) * \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix} \right]$$

$$= \Sigma \left[\begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} + \begin{bmatrix} -6 & -5 \\ -4 & -7 \end{bmatrix} \right]$$

$$= \Sigma \left[\begin{bmatrix} -6 & -3 \\ -2 & -4 \end{bmatrix} \right]$$

$$= -15$$

Again, $\Sigma a_i f_i(x, y) + \Sigma a_j f_j(x, y)$

$$= \Sigma \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} + \Sigma \begin{bmatrix} -6 & -5 \\ -4 & -7 \end{bmatrix}$$

$$= 7 - 22$$

$$= -15$$

Linear versus Nonlinear Operation

Two Properties of Linear Operation:

Additivity: $H[f_i(x, y) + f_j(x, y)] = H[f_i(x, y)] + H[f_j(x, y)]$

Homogeneity: $H[a_i f_i(x, y)] = a_i H[f_i(x, y)]$

- ❖ Equation indicates that the output of a linear equation due to the sum of two inputs is the same as performing the operation on the inputs individually and then summing the results.
- ❖ In addition, the output of a linear equation to a constant times an input is the same as the output of the operation due to original input multiplied by that constant.

Linear versus Nonlinear Operation

- ✓ Example: **Non-Linear Operation**
- ✓ Consider the max operation, whose function is to find the maximum value of pixels in an image,

$$\begin{aligned} \max [a_i f_i(x,y) + a_j f_j(x,y)] &\neq \max [a_i f_i(x,y)] + \max [a_j f_j(x,y)] \\ &\neq a_i \max [f_i(x,y)] + a_j \max [f_j(x,y)] \\ &\neq a_i g_i(x,y) + a_j g_j(x,y) \end{aligned}$$

$$\begin{aligned} \max \left\{ (1) \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} + (-1) \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix} \right\} &= \max \left\{ \begin{bmatrix} -6 & -3 \\ -2 & -4 \end{bmatrix} \right\} \\ &= -2 \end{aligned}$$

Working next with the right side, we obtain

$$\begin{aligned} (1) \max \left\{ \begin{bmatrix} 0 & 2 \\ 2 & 3 \end{bmatrix} \right\} + (-1) \max \left\{ \begin{bmatrix} 6 & 5 \\ 4 & 7 \end{bmatrix} \right\} &= 3 + (-1)7 \\ &= -4 \end{aligned}$$

Linear versus Nonlinear Operation

- ✓ Linear operation are exceptionally important, because they are based on a large body of theoretical and practical results that are applicable to image processing.
- ✓ Non-linear systems are not nearly as well understood, so there scope of application is more limited.
- ✓ But, there are several non-linear image processing operations whose performance far exceeds what is achievable by their linear counterparts.

Arithmetic Operations

❖ **Arithmetic operations** between images are array operations which, a means that arithmetic operations are carried out between corresponding pixel pairs.

$$s(x, y) = f(x, y) + g(x, y)$$

$$d(x, y) = f(x, y) - g(x, y)$$

$$p(x, y) = f(x, y) * g(x, y)$$

$$v(x, y) = f(x, y) / g(x, y)$$

❖ It is understood that, the operations that are performed between corresponding pixel pairs in f and g for $x=0, 1, 2, \dots, M-1$ and $y=0, 1, 2, \dots, N-1$. Where, M and N are the rows and columns of the images. Clearly, s , d , p and v are the images of $M \times N$ also.

Arithmetic Operation

Few important considerations:-

- ✓ If the result is a floating point number, round off it's value.
- ✓ If the result is above the pixel range, select the maximum range value.
- ✓ If the result is below the pixel range, select the minimum range value.
- ✓ If the result is infinity, write it as zero.

Arithmetic Operation

A			B			A+B			A			B			A*B		
$\begin{bmatrix} 0 & 100 & 10 \\ 4 & 0 & 10 \\ 8 & 0 & 5 \end{bmatrix}$	+	$\begin{bmatrix} 10 & 100 & 5 \\ 2 & 0 & 0 \\ 0 & 10 & 10 \end{bmatrix}$	=	$\begin{bmatrix} 10 & 200 & 15 \\ 6 & 0 & 10 \\ 8 & 10 & 15 \end{bmatrix}$	$\begin{bmatrix} 0 & 100 & 10 \\ 4 & 0 & 10 \\ 8 & 0 & 5 \end{bmatrix}$	*	$\begin{bmatrix} 10 & 100 & 5 \\ 2 & 0 & 0 \\ 0 & 10 & 10 \end{bmatrix}$	=	$\begin{bmatrix} 0 & 255 & 50 \\ 8 & 0 & 100 \\ 0 & 0 & 50 \end{bmatrix}$								

A			B			A-B			A			B			A/B		
$\begin{bmatrix} 0 & 100 & 10 \\ 4 & 0 & 10 \\ 8 & 0 & 5 \end{bmatrix}$	-	$\begin{bmatrix} 10 & 100 & 5 \\ 2 & 0 & 0 \\ 0 & 10 & 10 \end{bmatrix}$	=	$\begin{bmatrix} 0 & 0 & 5 \\ 2 & 0 & 10 \\ 8 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 100 & 10 \\ 4 & 0 & 10 \\ 8 & 0 & 5 \end{bmatrix}$	/	$\begin{bmatrix} 10 & 100 & 5 \\ 2 & 0 & 0 \\ 0 & 10 & 10 \end{bmatrix}$	=	$\begin{bmatrix} 0 & 1 & 2 \\ 2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$								

Here, all the considered images are of gray level
and their intensity value ranges from 0.....255

Arithmetic Operation (Addition)

- ✓ **Example:** Addition (averaging) of noisy images for noise reduction.
- ✓ Used in **image enhancement**.
- ✓ **Image averaging is used in the field of astronomy.**
- ✓ Let $g(x,y)$ denote a corrupted image formed by the addition of noise, $\eta(x, y)$ to a noiseless image $f(x,y)$; that is,

$$g(x,y) = f(x,y) + \eta(x,y)$$

- ✓ The objective of the following procedure is to reduce the noise content of the output image by adding a set of noisy input images, This is a technique used frequently for image enhancement.

Arithmetic Operation (Addition)

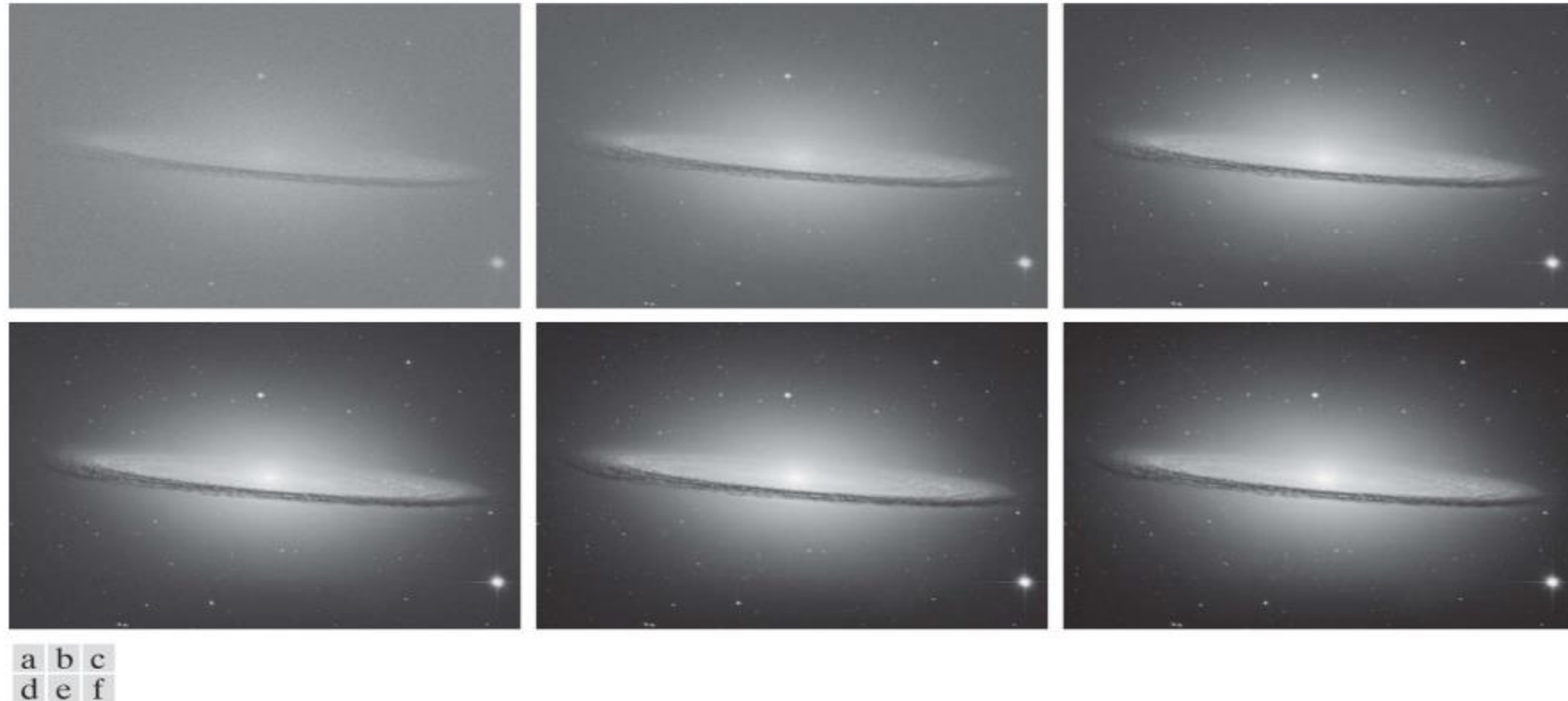


Figure 1: (a) Sample noisy image of the Sombrero Galaxy. (b)-(f) Result of averaging 10, 50, 100, 500, and 1,000 noisy images, respectively. All images are of size pixels, and all were scaled so that their intensities would span the full $[0, 255]$ intensity scale.

Arithmetic Operation

The mean of a sample is

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

Mean does not tell us a lot about the data except for a sort of middle point.

X = [0 8 12 20]
Mean = 10

Y = [8 9 11 12]
Mean = 10

What is different about two sets?

Arithmetic Operation

The standard deviation of a sample is,

$$\text{Standard deviation, } \sigma = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n-1}}$$

Why are we using n-1 and not n?

Because we take a subset of the real world and this value represents the closer to the standard deviation. If we can take full sample then we can use n instead of n-1.

Arithmetic Operation

What is different about two sets?

$X = [0 \ 8 \ 12 \ 20]$
Mean – 10
Standard Deviation ~ 8.33

$Y = [8 \ 9 \ 11 \ 12]$
Mean – 10
Standard Deviation ~ 1.83

Result?

Better

Arithmetic Operation (Subtraction)

- ✓ A frequent application of **image subtraction** is in the enhancement of including correcting uneven illuminance and comparing two images.
- ✓ Mask mode radiography in medical imaging.

Arithmetic Operation (Subtraction)



Figure 2: (a) Original Image (b) Reference Image
(c) Difference of the two images, scaled to the range $[0, 255]$ for clarity.

Arithmetic Operation (Subtraction)

- ✓ Image subtraction in medical imaging is called *mask mode radiography*

$$g(x, y) = f(x, y) - h(x, y)$$

- ✓ In this case **$h(x, y)$** , **the mask**, is an X-ray image of a region of a patient's body captured by an intensified TV camera (instead of traditional X-ray film) located opposite an X-ray source.
- ✓ The procedure consists of injecting an X-ray contrast material into the patient's bloodstream, taking a series of images called *live images*, **$f(x, y)$** , samples of which are denoted as $f(x, y)$ of the same anatomical region as $h(x, y)$, and subtracting the mask from the series of incoming live images after injection of the contrast material.
- ✓ The net effect of subtracting the mask from each sample live image is that the areas that are different between $f(x, y)$ and $h(x, y)$ appear in the output image, $g(x, y)$, as enhanced detail.

Arithmetic Operation (Subtraction)

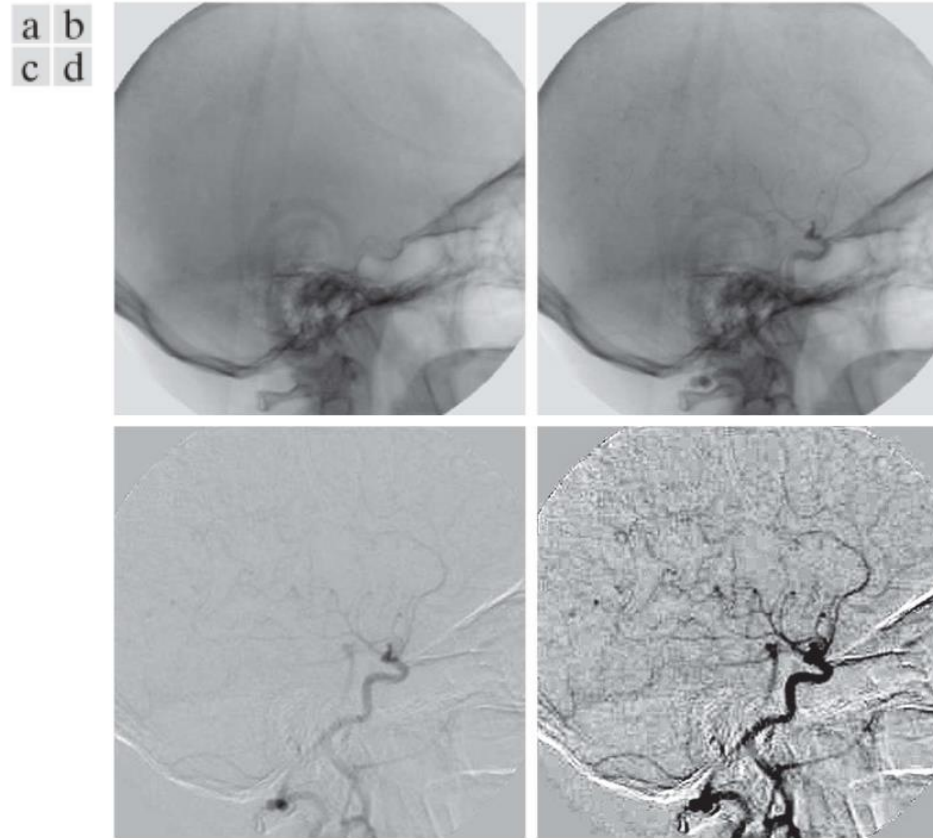


Figure 3: Digital subtraction angiography. (a) Mask image. (b) A live image. (c) Difference between (b) and (a). (d) Enhanced difference image.

Arithmetic Operation (Multiplication and Division)

- ❖ An important application of image **multiplication (and division)** is *shading correction*.
- ❖ Suppose that an imaging sensor produces images, $g(x,y)$,

$$g(x,y) = f(x,y)h(x,y)$$

- $f(x,y)$ indicates **perfect image**

- $h(x,y)$ indicates **a shading function**

Arithmetic Operation (Multiplication and Division)

If $h(x, y)$ is known!!!!

$$f(x, y) = g(x, y) \div h(x, y)$$

- multiplying the sensed image $g(x, y)$ by the **inverse** of $h(x, y)$

Arithmetic Operation (Multiplication and Division)

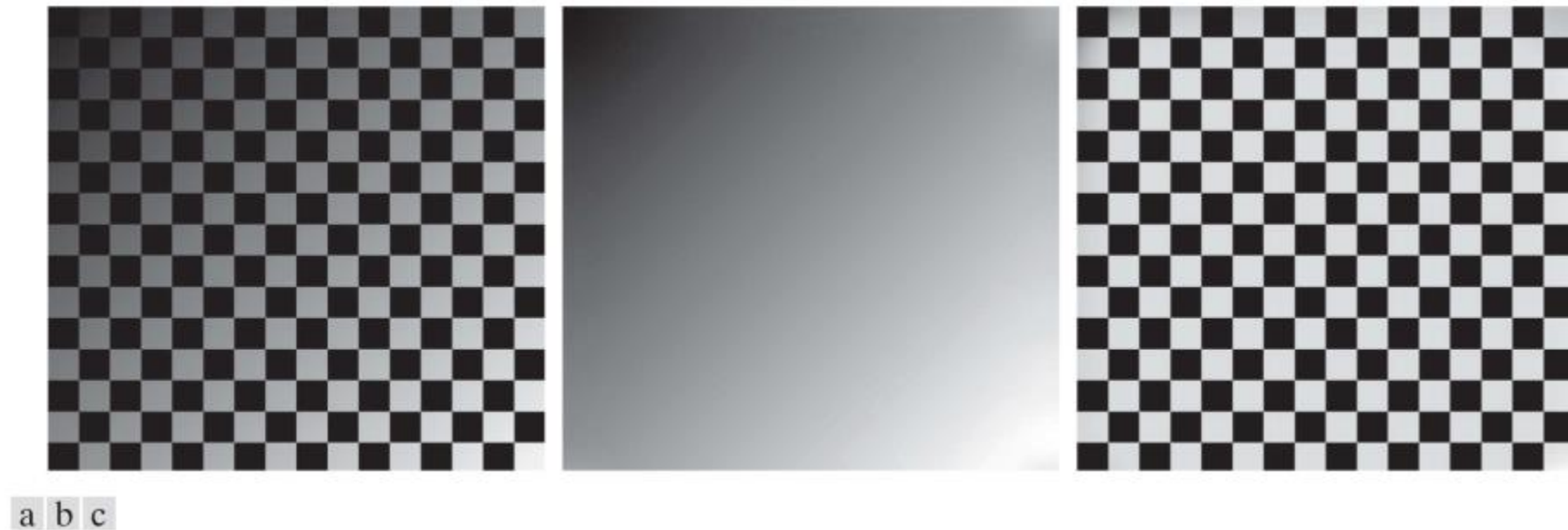


Figure 4: Shading correction. (a) Shaded test pattern. (b) Estimated shading pattern.
(c) Product of (a) by the reciprocal of (b).

Arithmetic Operation (Multiplication and Division)

- ✓ Another common use of image multiplication is in **masking**, also called **region of interest (ROI)**, operations.
- ✓ Multiplying a given image by a **mask** image that **has 1's in the ROI** and 0's elsewhere.



Figure 5: (a) Digital dental X-ray image. (b) ROI mask for isolating teeth with fillings (white corresponds to 1 and black corresponds to 0). (c) Product of (a) and (b).

Set Operations

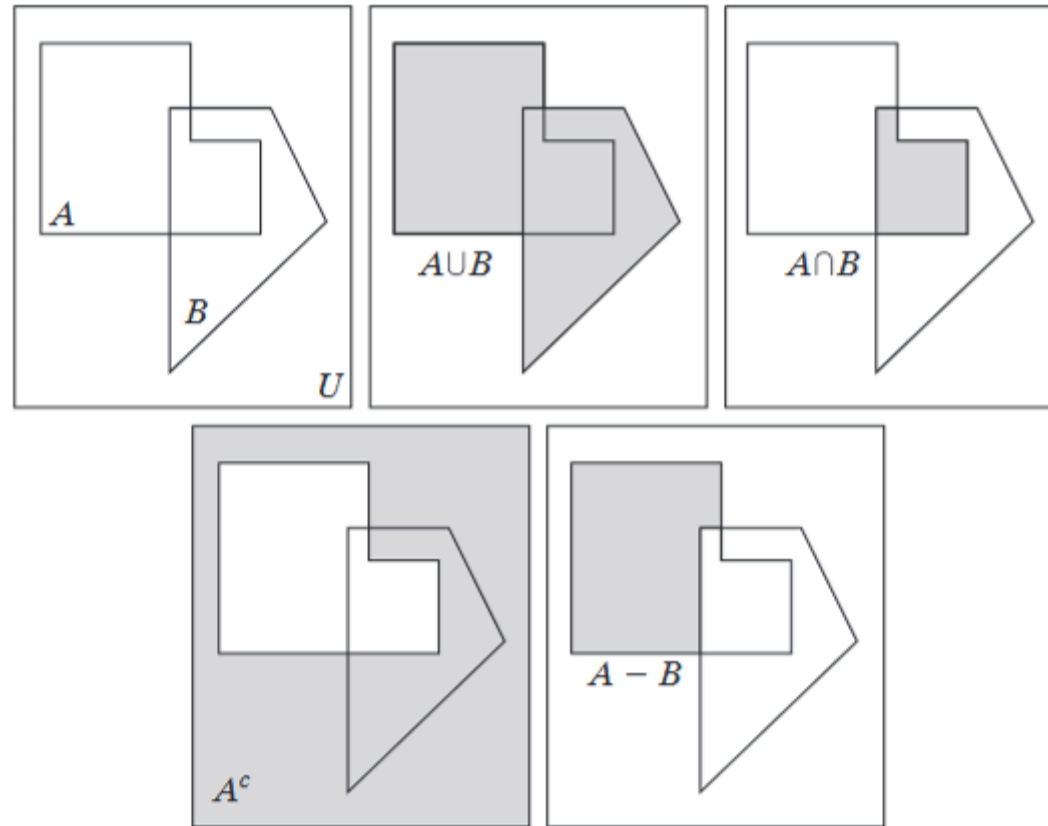
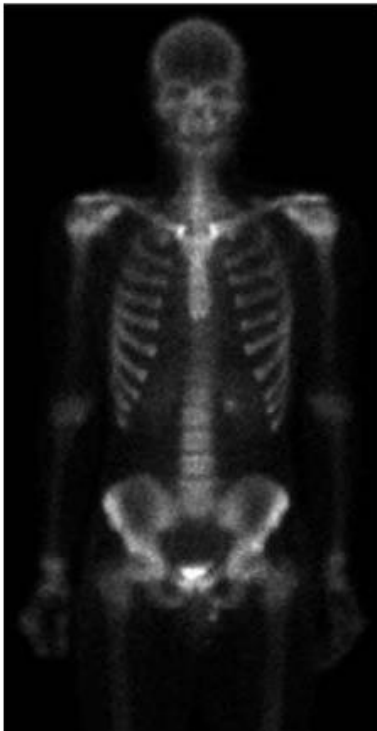


Image Negatives

- ✓ **Set** operations involving image **intensities**.
- ✓ Let the elements of a grayscale image be represented by a set A whose elements are triplets of the form (x, y, z) , where x and y are spatial coordinates, and z denotes intensity values. We define the *complement* of A as the set,

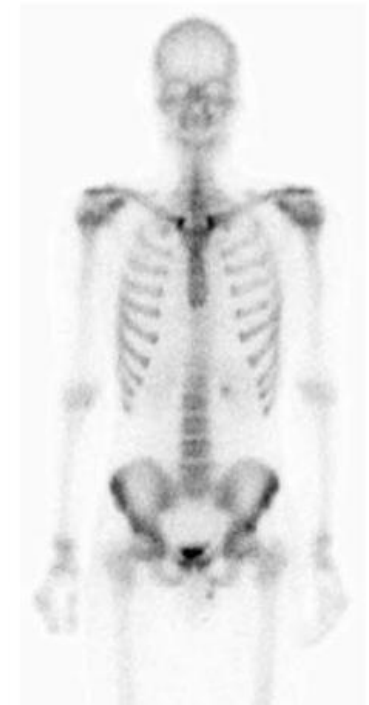
$A =$



$$A^c = \{(x, y, K - z) \mid (x, y, z) \in A\}$$

$$K = 2^k - 1, \quad k - \text{number of bits}$$

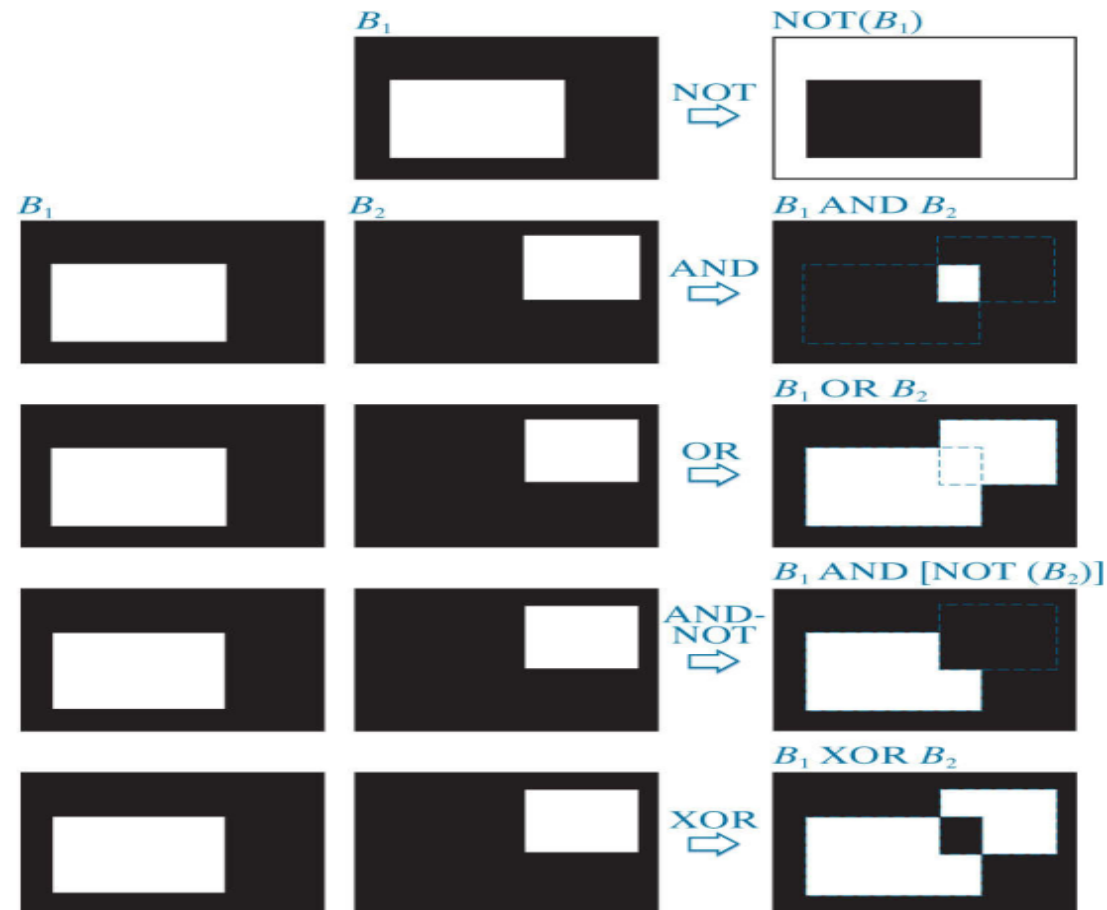
$A^c =$



Logical Operations

- ✓ Logical operations deal with TRUE (typically denoted by 1) and FALSE (typically denoted by 0) variables and expressions.
- ✓ When dealing with binary images, we can think of **foreground (1-valued)** and **background (0-valued)** sets of pixels.
- ✓ We work with set and logical operators on binary images using one of two basic approaches: (1) we can use the *coordinates* of individual regions of foreground pixels in a single image as sets, or (2) we can work with one or more images of the same size and perform logical operations between corresponding pixels in those arrays.
- ✓ When dealing with binary images, it is common practice to refer to **union**, **intersection**, and **complement** as the **OR**, **AND**, and **NOT** logical operations, where “logical” arises from logic theory in which 1 and 0 denote true and false, respectively.

Logical Operations



Spatial Operations

- ✓ Spatial operations are performed directly on the pixels of a given image.
- ✓ Classifications of spatial operations:
 - 1. Single pixel operations**
 - 2. Neighborhood operations**
 - 3. Geometric spatial transformations**

Spatial Operations

✓ Single Pixel Operations

- Alter the values of an image pixels based on intensity.
- Finding the negative of a given image.
- Example: **NOT** operation.

0	0	0	0
0	1	1	0
0	1	1	0
0	1	1	0
0	0	0	0



1	1	1	1
1	0	0	1
1	0	0	1
1	0	0	1
1	1	1	1

Spatial Operations

✓ Single Pixel Operations

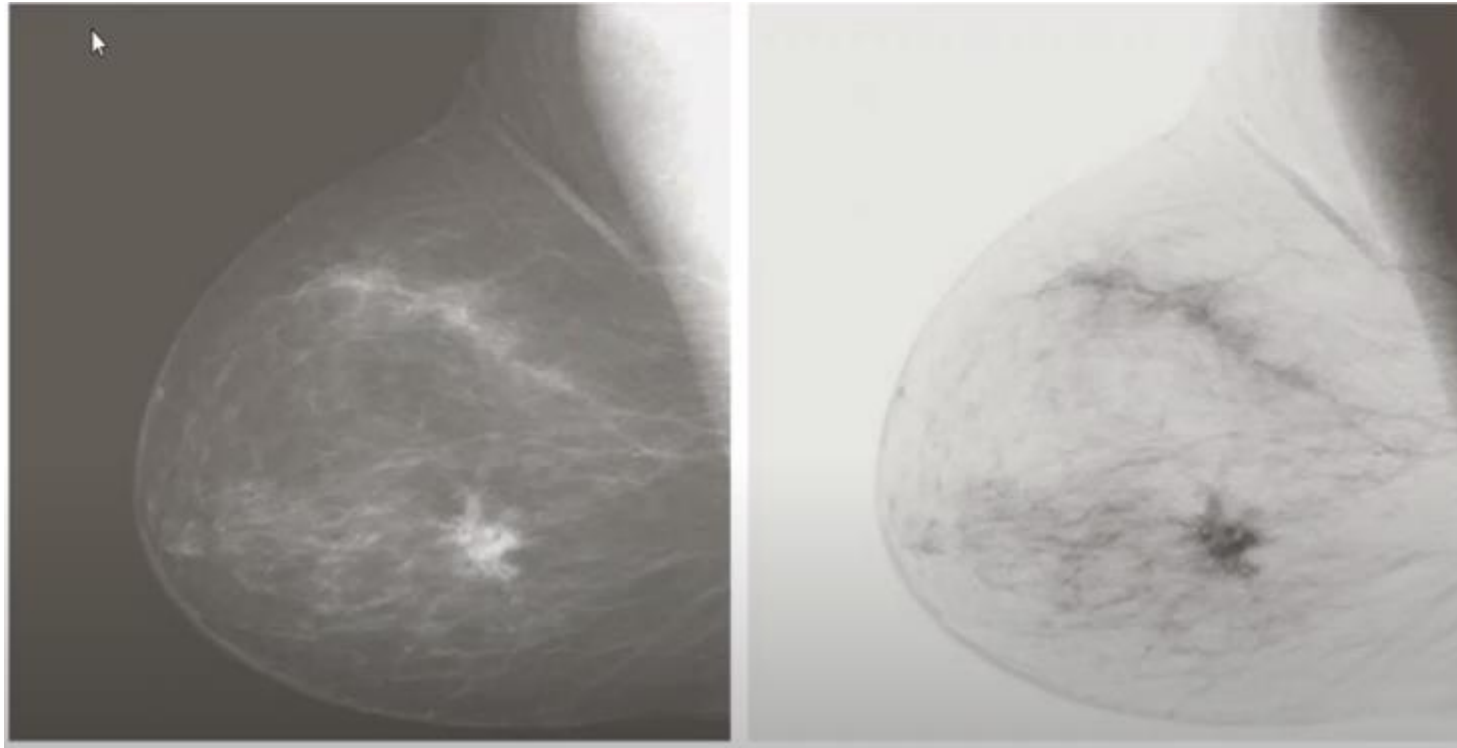


Figure 6: Image Negatives

Spatial Operations

✓ Neighborhood Operations

- The net effect is to perform **local blurring** in the original image.

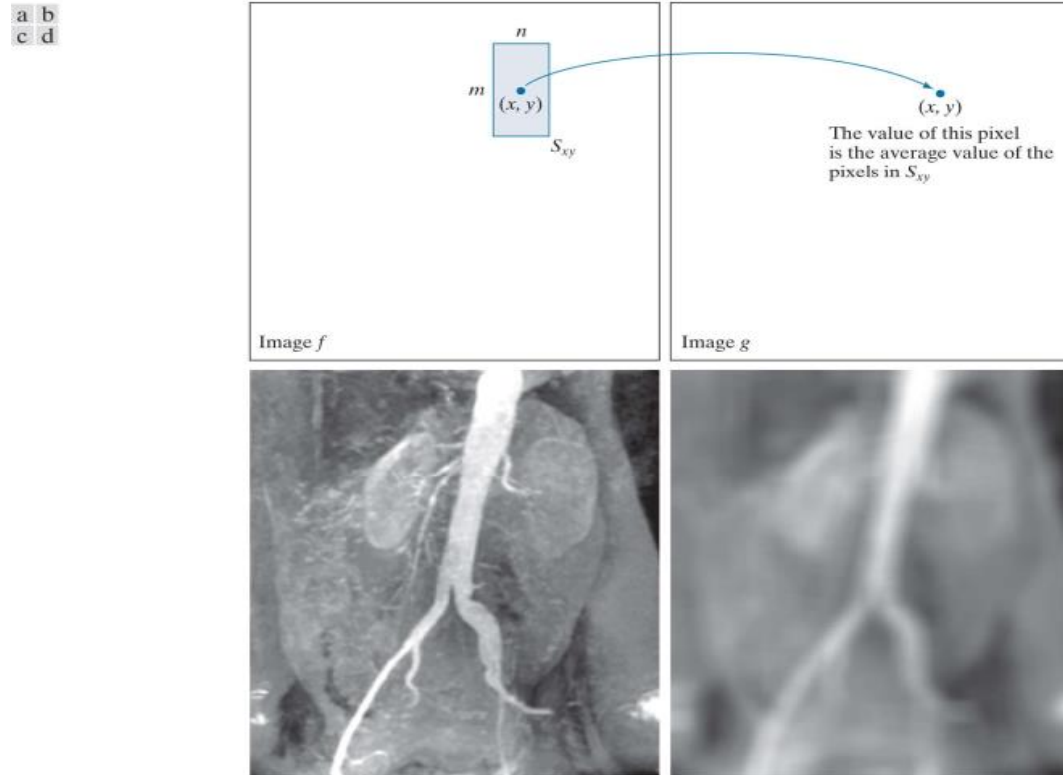


Figure 6: Local averaging using neighborhood processing. The procedure is illustrated in (a) and (b) for a rectangular neighborhood. (c) An aortic angiogram (d) The result of using neighborhood operations.

Spatial Operations

✓ Neighborhood operations

At first we need to select the center pixel and find its 3x3 neighbors and calculate the average of all neighbor pixels and update the center pixel.

7	5	2	7	2	1	5
1	5	2	9	2	8	8
7	4	1	8	1	4	5
6	4	2	7	2	2	5
6	5	2	4	2	1	5
5	5	2	5	2	8	4
5	4	2	2	2	5	5
5	4	3	4	3	6	2

Spatial Operations

✓ Neighborhood operations

At first we need to select the border pixel as the center pixel and to find it's 3x3 neighbors, zero padding is needed and calculate the average of all neighbor pixels and update the center pixel.

0	0	0	0	0	0	0	0	0
0	7	5	2	7	2	1	5	0
0	1	5	2	9	2	8	8	0
0	7	4	1	8	1	4	5	0
0	6	4	2	7	2	2	5	0
0	6	5	2	4	2	1	5	0
0	5	5	2	5	2	8	4	0
0	5	4	2	2	2	5	5	0
0	5	4	3	4	3	6	2	0
0	0	0	0	0	0	0	0	0

Spatial Operations

✓ Geometric spatial transformations

-Spatial transformation of an image is a geometric transformation of the image coordinate system.

$$(v, w) = T\{(x, y)\}$$

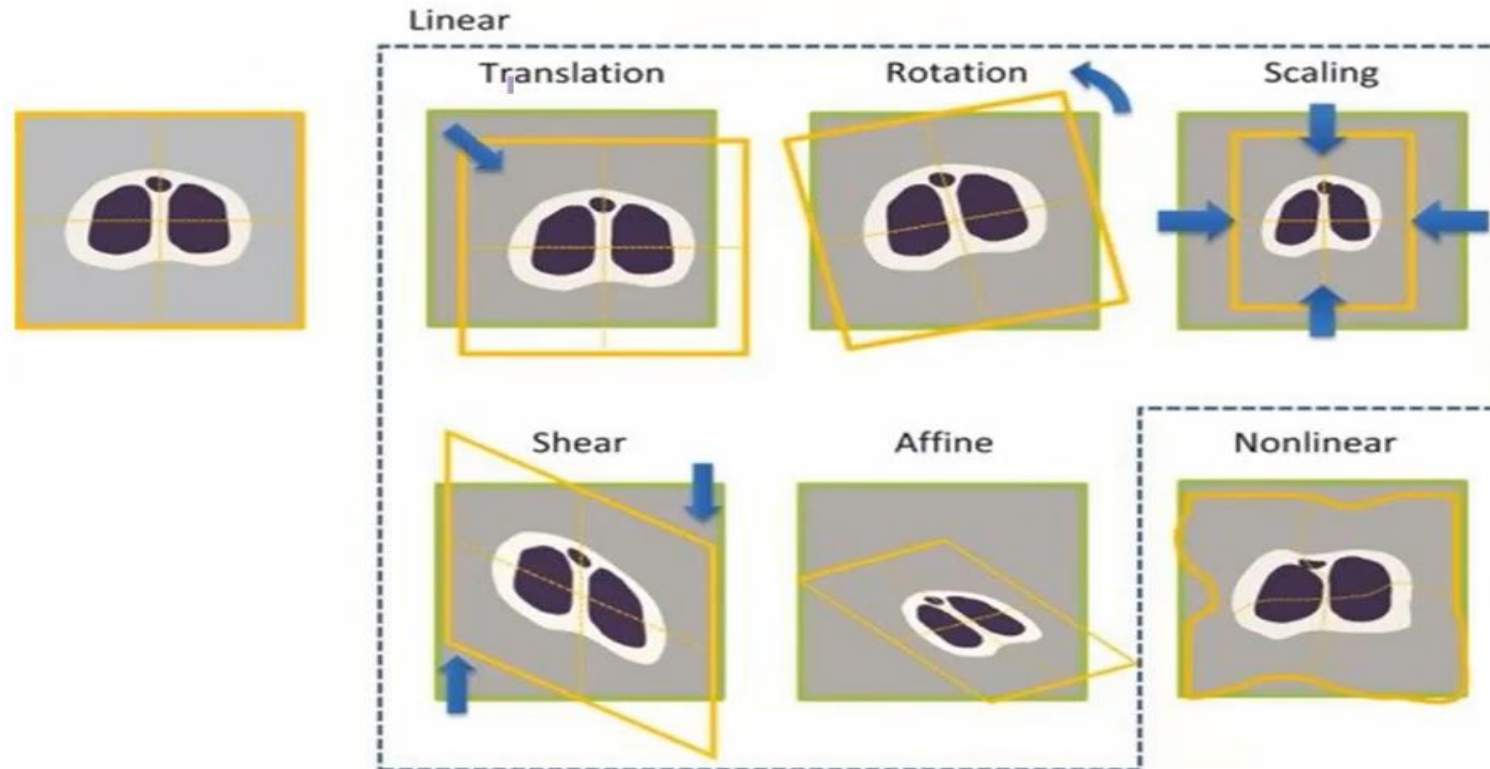
Where,

(x, y) - pixel coordinates in the **original** image

(v, w) -The corresponding pixel coordinates in the **transformed** image

Spatial Operations

✓ Geometric spatial transformations



Spatial Operations

❖ Geometric Spatial Operations (Translation)

✓ It is known as shifting of object's location.

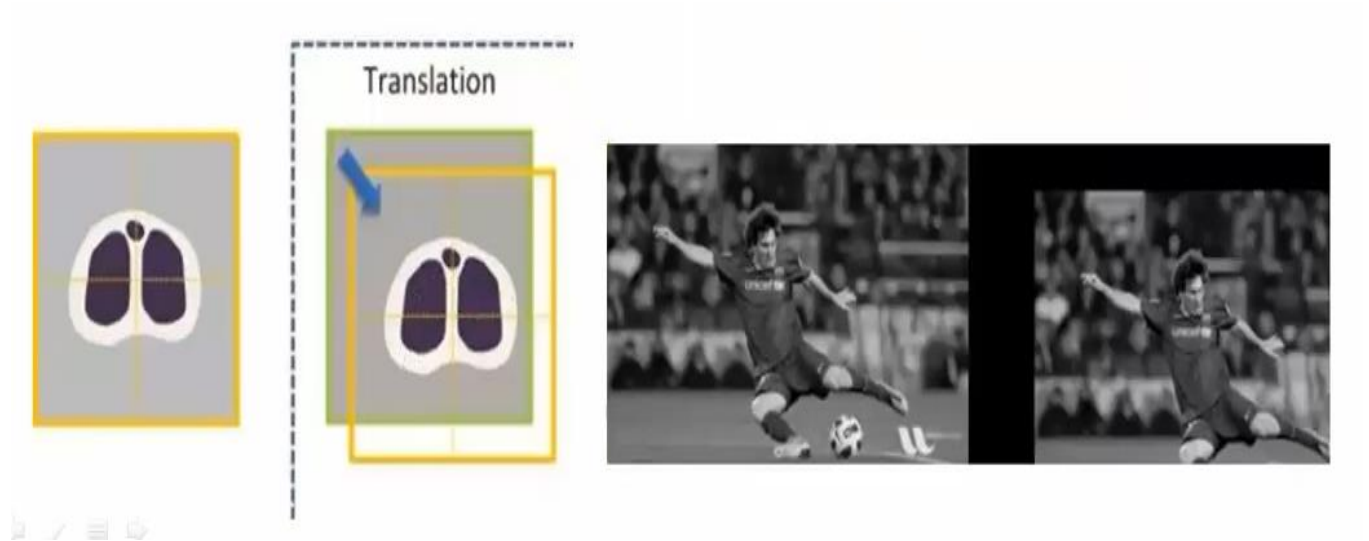
✓ Here, $(x, y) = (0, 0)$

✓ $(t_x, t_y) = (15, 13)$

Here,

$$M = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



Spatial Operations

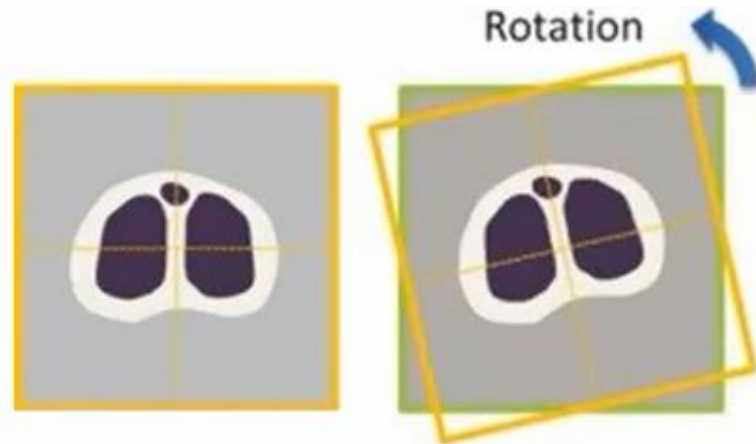
✓ Geometric spatial transformations (Rotation)

- It rotates an image by a specified angle.

- Here,

- $$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

- $(x, y) = (20, 40)$

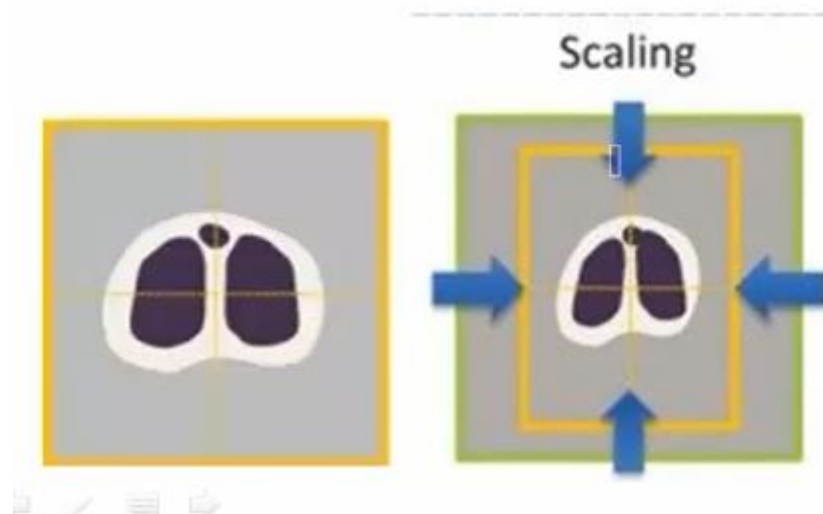


Spatial Operations

✓ Geometric spatial transformations (Scaling)

- Scaling means resize of an image in terms of scaling factor.
- If we have an image of (300 X 400) and we want to transform it into a shape of (600 X 200).
- The scaling in X and Y direction will be $600/300=2$ and $200/400 = 1/2$

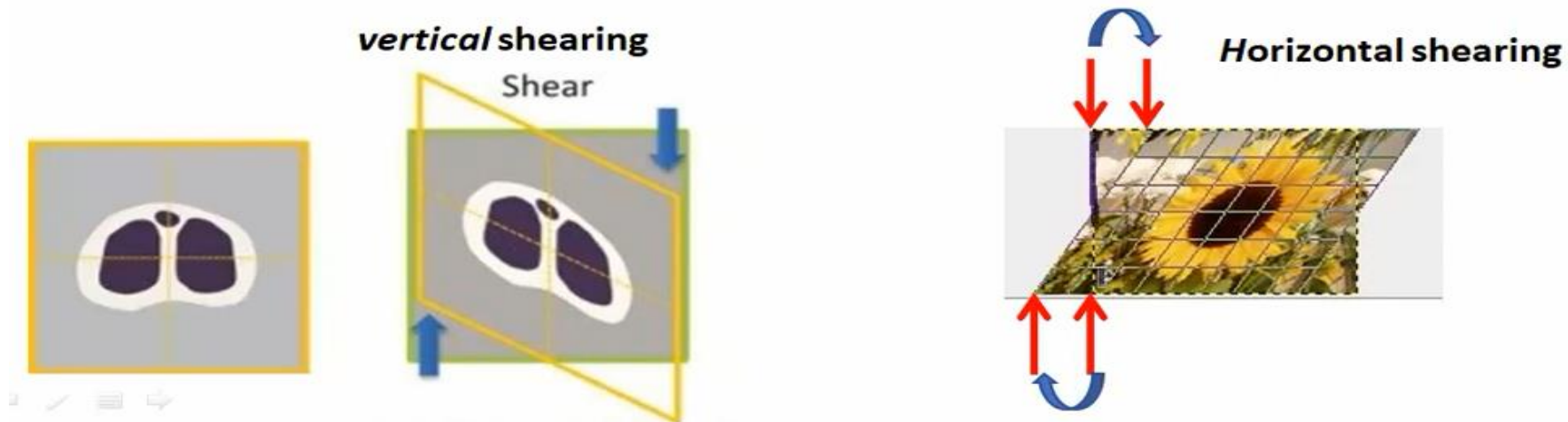
$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix}$$



Spatial Operations

✓ Geometric spatial transformations (Shearing)

- Shearing an image means shifting the pixel values either horizontally or vertically.
- Basically, this shifts some part of an image to one direction and other parts to the another direction.
Horizontal shearing will shift the upper part to the right and lower part to the left. Vertical shearing will shift left part to upper and right part to lower.



Spatial Operations

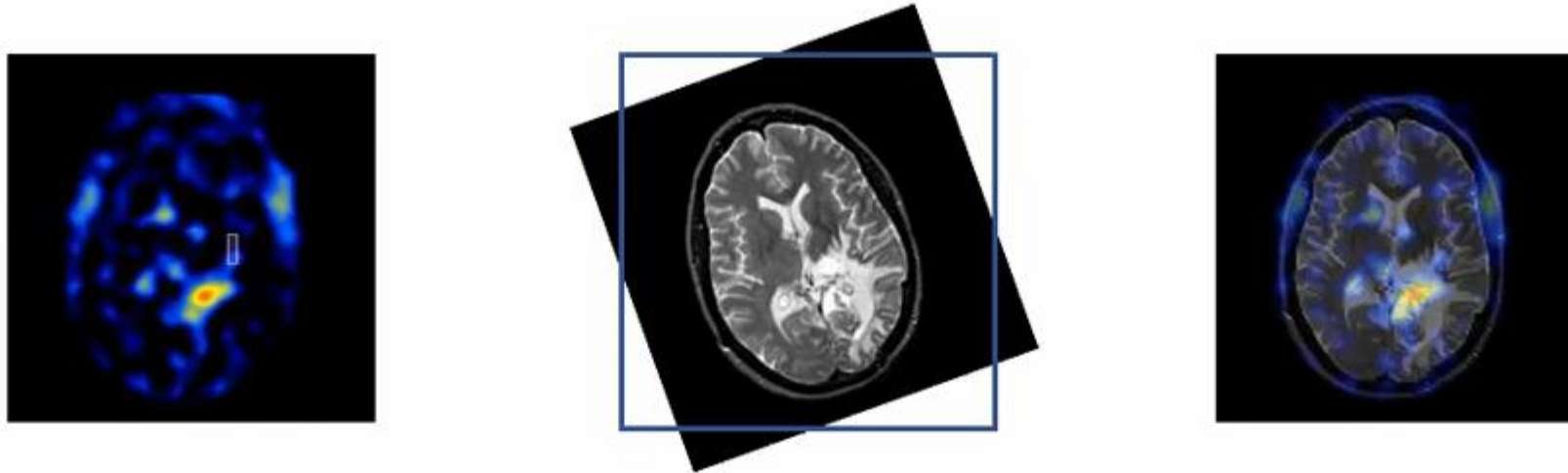
❖ Geometric spatial transformations and image registration

- Image registration is the process of transforming different sets of data into one co-ordinate system.
- Image registration is an important application of digital image processing used to align two or more images of the same scene. In image registration, we have available an *input* image and a *reference* image. The objective is to transform the input image geometrically to produce an output image that is aligned (registered) with the reference image.
- **Application:** Fusion of medical images, like SPECT-MRI, CT-MRI etc.

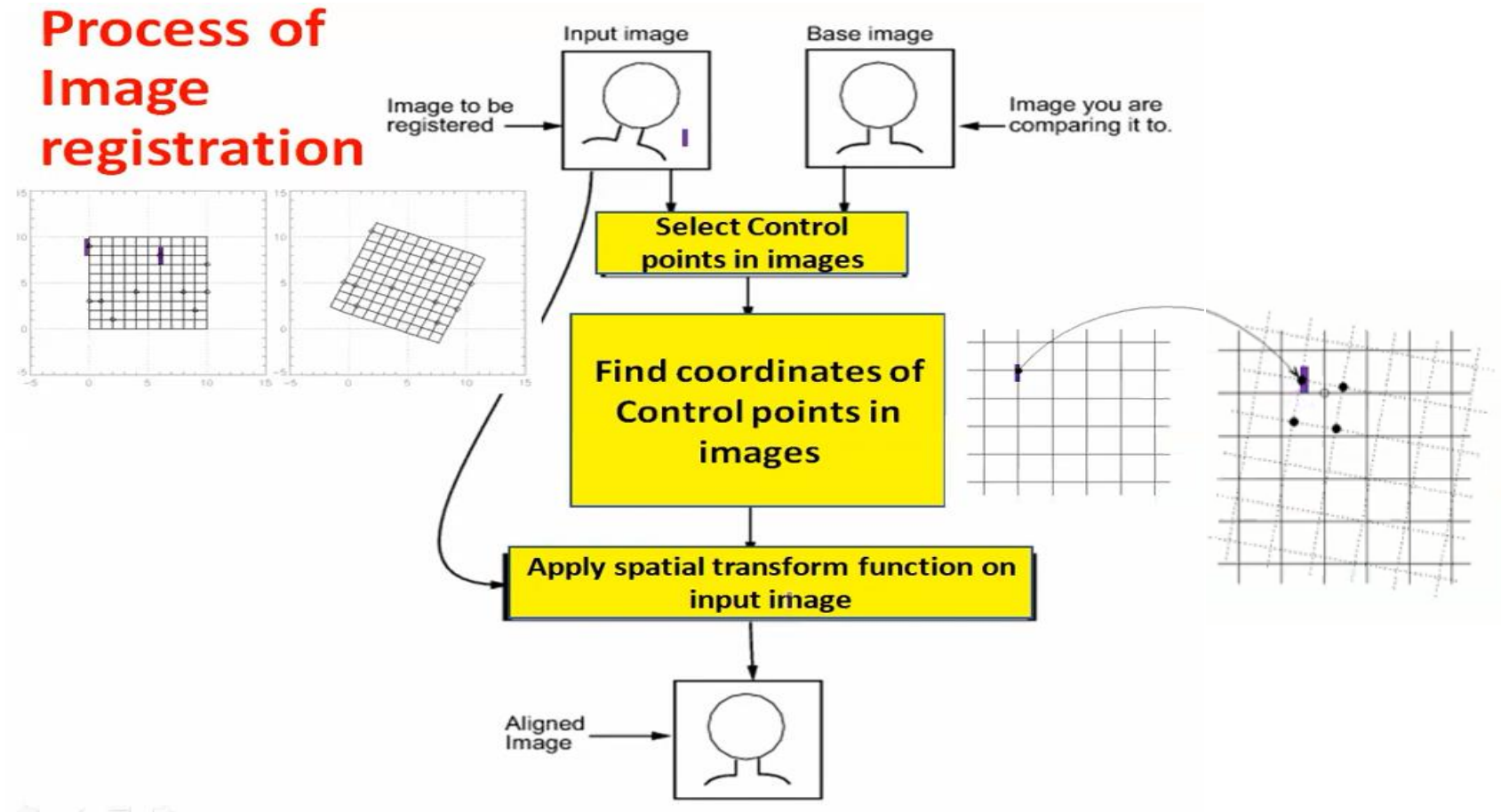
Spatial Operations

❖ Geometric spatial transformations and image registration

- In many applications, it is necessary to combine multiple images of the same scene acquired by different sensors or same sensors, from same or different viewpoint.
- Following the figures indicate SPECT (Single-Photon Emission Computerized Tomography) and MRI images of brain.



Spatial Operations



Thank You 😊